

Summary

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0.1 How to Solve an Optimization Problem

- (a) Identify **variables**; draw and label the picture of the problem
- (b) Identify the **objective function** (a quantity to be optimized); write a **formula** for the objective function in terms of variables of the problem
- (c) Identify the **constraint(s)**; use the constraint(s) to express all the variables in terms of a **single variable**
- (d) Write the **objective function in terms of a single variable**; find the **interval of interest**
- (e) Using the methods of calculus, find the **global maximum/minimum**; **justify** your answer

REMINDER: The Interval of Interest tells you the method to use to solve an optimization problem.

CASE 1: The interval of interest is a closed interval $[a, b]$

In this case, the Extreme Value Theorem (EVT) guarantees that both global extrema of f exist!

If f has an global minimum and an global maximum which either occur at a critical point or at a boundary point (which means a or b). To find an extreme values of f on $[a, b]$, we:

- Find all critical points and plug them into f .
- Evaluate f at both boundary points.
- Compare the values. The biggest of those values is the maximum value of f on $[a, b]$, and the least one is the minimum value of f on $[a, b]$.

CASE 2: The interval of interest is any interval.

In this case, the global minimum/maximum **may not exist**. Our approach is then:

- Find all critical points of f on the interval.
- Use 1st or 2nd Derivative Test to classify those critical points as **local maxima/minima**.
- If there is *exactly one local extremum*, then it is an global extremum of the same type. (That is, if it is a local minimum then it is automatically a global minimum...)

OTHER CASES (e.g., An interval with multiple local extrema, etc) can be approached similar to how we graphed functions: check the boundary points, check the critical points, use the sign chart etc.